

Data Structures for Post Interaction Handler

1. Credentials

Credentials = Username + Password

Username = string

Password = string

2. Authentication Response

Authentication_Response = Auth_Status + Session_Token + Error_Message

Auth_Status = {Success | Failure}

Session_Token = string (for authenticated users)

Error_Message = string (if authentication fails)

3. Home Feed

Home_Feed = Recent_Posts + Notification_Updates + User_Interactions

Recent_Posts = {Post_ID + Post_Title + Post_Content + Post_Author}

Notification_Updates = {New_Comment | New_Vote | Post_Update}

User_Interactions = {Like | Dislike | Comment}

4. New Post

New_Post = Post_Title + Post_Content + Author_ID + Post_Timestamp + Media_Attachments +
Post_Status

Post_Title = string

Post_Content = string

Author_ID = string

Post_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

Media_Attachments = {Image | Video | None}

Post_Status = {Draft | Published | Pending_Approval}

5. Comment

Comment = Comment_Text + Comment_Timestamp + Post_ID + Author_ID

Comment_Text = string

Comment_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

Post_ID = string

Author_ID = string

6. Cast Votes

Cast_Votes = Vote_Type + Post_ID + User_ID

Vote_Type = {Upvote | Downvote}

Post_ID = string

User_ID = string

7. Search Subforum

Search_Subforum = Search_Query + Filter_Options

Search_Query = string

Filter_Options = {Date_Filter | Category_Filter | Author_Filter}

8. Update Alert

Update_Alert = Post_ID + Update_Type + Timestamp + New_Content

Post_ID = string

Update_Type = {Comment | Vote | Status_Change}

Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

New_Content = string

9. Approval Request

Approval_Request = Post_ID + Moderator_ID + Approval_Status

Post_ID = string

Moderator_ID = string

Approval_Status = {Pending | Approved | Rejected}

10. Approval Confirmation

Approval_Confirmation = Post_ID + Approval_Status + Timestamp

Post_ID = string

Approval_Status = {Approved | Rejected}

Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

11. Load Views

Load_Views = Post_ID + View_Count + Timestamp

Post_ID = string

View_Count = integer

Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

12. Moderation Views

Moderation_Views = Post_ID + Moderator_ID + Action_Taken

Post_ID = string

Moderator_ID = string

Action_Taken = {Approved | Rejected | Flagged}

13. Moderation Alert

Moderation_Alert = Post_ID + Flagged_By + Reason + Alert_Timestamp

Post_ID = string

Flagged_By = string

Reason = string

Alert_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

14. Authentication Request (to server)

Authentication_Request = Username + Password + Request_Timestamp

Username = string

Password = string

Request_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

15. Confirmation (From Server)

Confirmation = Auth_Status + Session_Token + Error_Message

Auth_Status = {Success | Failure}

Session_Token = string (for authenticated users)

Error_Message = string (if authentication fails)

16. Post View

Post_View = Post_ID + Viewer_ID + View_Timestamp

Post_ID = string

Viewer_ID = string

View_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

17. Get Posts

Get_Posts = Subforum_ID + Post_List + Query_Timestamp

Subforum_ID = string

Post_List = {Post_ID + Post_Title + Post_Author}

Query_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

18. Query Response

Query_Response = Post_List + Query_Timestamp

Post_List = {Post_ID + Post_Title + Post_Author}

Query_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

19. Fetch Recent Posts

Fetch_Recent_Posts = Post_List + Query_Timestamp

Post_List = {Post_ID + Post_Title + Post_Author}

Query_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

20. Recent Posts

Recent_Posts = {Post_ID + Post_Title + Post_Author + Post_Timestamp}

Post_ID = string

Post_Title = string

Post_Author = string

Post_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

21. Action History

Action_History = Action_Type + Post_ID + Timestamp + Moderator_ID

Action_Type = {Approved | Rejected | Flagged}

Post_ID = string

Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

Moderator_ID = string

22. Post Update

Post_Update = Post_ID + Updated_Content + Update_Timestamp

Post_ID = string

Updated_Content = string

Update_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

23. Post Approval

Post_Approval = Post_ID + Approval_Status + Moderator_ID

Post_ID = string

Approval_Status = {Approved | Rejected}

Moderator_ID = string

24. Get Notification Preferences

Get_Notification_Preferences = User_ID + Notification_Preferences

User_ID = string

Notification_Preferences = {Email | SMS | Web | None}

25. Notification Preferences

Notification_Preferences = {Email | SMS | Web | None}

26. Log Notification

Log_Notification = Notification_ID + User_ID + Timestamp + Notification_Type

Notification_ID = string

User_ID = string

Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

Notification_Type = {Post_Interaction | System_Alert}

27. Store Post

Store_Post = Post_Content + Post_Timestamp + Author_ID

Post_Content = string

Post_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

Author_ID = string

28. Update Post

Update_Post = Post_ID + Updated_Content + Update_Timestamp

Post_ID = string

Updated_Content = string

Update_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")

29. Search Query to D2

Search_Query_To_D2 = Search_Query + Filter_Options

Search_Query = string

Filter_Options = {Date_Filter | Category_Filter | Author_Filter}

30. Load Feed

Load_Feed = Post_List + User_ID + Feed_Timestamp

Post_List = {Post_ID + Post_Title + Post_Author}

User_ID = string

Feed_Timestamp = datetime ("YYYY-MM-DD HH:MM:SS")